

# Dimensionality Reduction Principal Component Analysis (PCA)

 Professor: M. Sc. Héctor A. Hormazábal Vildósola

## MOTIVATION - DIMENSIONALITY REDUCTION

In Machine Learning (ML) and statistics, we often work with datasets containing many different features or variables, typically denoted as  $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \dots, \mathbf{X}_n$ . While having more data sounds better, high-dimensional datasets often come with significant challenges. As we add more variables, the model becomes increasingly complex, significantly raising the risk of overfitting; the model tends to memorize the specific noise of the training data rather than generalizing the intrinsic patterns, leading to poor performance on testing sets.

In this context, *dimensionality reduction* serves as a fundamental step to distill the dataset into its most informative essence. By projecting the high-dimensional observations onto a lower-dimensional manifold, we aim to uncover the latent structure that governs the data generation process. Mathematically, if our original dataset  $\mathbf{X} \in \mathbb{R}^{m \times n}$  is a matrix ( $m$  - rows-registers  $\times$   $n$  - columns-features-variables), we seek a transformation into a reduced space  $\mathbb{R}^{m \times k}$  (where  $k \ll n$ ) that preserves the **maximum amount of information - variance**.

This process not only improves computational efficiency but also acts as a powerful regularizer, forcing the model to ignore redundant correlations and noise. Furthermore, effective reduction facilitates the visualization of complex relationships that are otherwise impossible to perceive in  $n$ -dimensional space.

## THE BUSINESS ANALYST'S PERSPECTIVE: DRIVING STRATEGIC CLARITY

For a **Business Intelligent (BI)** Analyst, high-dimensional data often translates into "information noise", where too many variables obscure the key drivers of performance. Dimensionality reduction is a powerful tool for **Strategic Simplification**. By distilling complex datasets into their most impactful components, analysts can identify the underlying patterns that truly influence customer behavior, market trends, or operational risks.

In practice, this allows for more robust customer segmentation and predictive modeling without the risk of overfitting. Instead of juggling dozens of correlated metrics, the analyst focuses on the "latent signals" that represent the business's core reality. This process not only enhances model accuracy but also facilitates **Data Storytelling**, turning dense technical outputs into clear, actionable insights.

## Case Study: Optimization of Customer Credit Risk Assessment

In the financial services sector, assessing credit risk is a high-dimensional challenge that requires analyzing hundreds of behavioral and demographic data points. For a **Business Analyst**, the primary objective is to distinguish between high-value clients and potential defaulters without getting lost in redundant metrics. Dimensionality reduction allows the organization to condense these numerous variables into a handful of "Risk Indicators," improving the predictive power of credit scoring models and streamlining the approval process.

**The Strategic Advantage:** By applying **advanced dimensionality reduction techniques** in this context, the business achieves significant **operational efficiency** and analytical precision. Instead of processing a massive matrix of sparse, overwhelming data, the analyst can work with synthesized features that capture the true variance and underlying signals in repayment behavior. This reduction minimizes the "statistical noise" caused by temporary market fluctuations or redundant indicators, allowing the organization to focus exclusively on the structural financial health and creditworthiness of the applicant. Furthermore, this streamlined approach accelerates the time-to-insight, enabling faster credit decisions that directly improve the customer experience. This structured approach enables a more **transparent communication** with executive stakeholders, as the model's logic is based on clear, interpreted dimensions rather than an incomprehensible list of raw variables.

Business Case: Vanguard Capital Group

Vanguard Capital Group needs to optimize its risk engine for 15,000 high-profile clients. While real-world financial models often involve hundreds of features, the challenge here is illustrated by variables  $\mathbf{X_1}$  through  $\mathbf{X_6}$ . This serves to demonstrate how excessive dimensionality creates a "dispersion" of information—the *curse of dimensionality*—making it difficult to find clear, actionable patterns.

The primary objective is to apply data reduction to simplify these variables into a few "dimensions" that are easy to interpret. By filtering analytical noise, specialists can extract strategic insights into solvency and engagement drivers, rather than getting lost in isolated data points.

This approach enables Vanguard’s analysts to transition from technical descriptions to executive decision-making, identifying risk groups and market opportunities in a robust and visual manner.

Table 1: Key Variable Clusters for Credit Assessment

| Variable | Dimension           | Description                                                                           | Units          | Type                    |
|----------|---------------------|---------------------------------------------------------------------------------------|----------------|-------------------------|
| $X_1$    | Liquidity Profile   | Average monthly balance in liquid accounts over the last quarter.                     | USD (\$)       | Quantitative Continuous |
| $X_2$    | Credit History      | Number of months with at least one payment delay in the last 2 years.                 | Months ( $n$ ) | Quantitative Discrete   |
| $X_3$    | Spending Stability  | Standard deviation of monthly discretionary expenditures during the last fiscal year. | $\sigma$ (USD) | Quantitative Continuous |
| $X_4$    | Asset Solvency      | Ratio of total verified assets to outstanding long-term liabilities.                  | Ratio (%)      | Quantitative Continuous |
| $X_5$    | Demographic Risk    | Total number of years in the current professional position or industry.               | Years ( $t$ )  | Quantitative Discrete   |
| $X_6$    | Behavioral Patterns | Frequency of digital platform logins per month as a proxy for engagement.             | Logins / Month | Quantitative Discrete   |

Principal Component Analysis

Principal Component Analysis (PCA) is a fundamental **unsupervised machine learning algorithm** used for **dimensionality reduction**. Its primary objective is to transform a high-dimensional space into a lower-dimensional representation while preserving as much of the data’s original variance as possible (**maximize the explained variance by the principal components**). For an Industrial Engineering perspective, this process is essentially a **constrained optimization problem** that simplifies complex business datasets into their most informative essence.

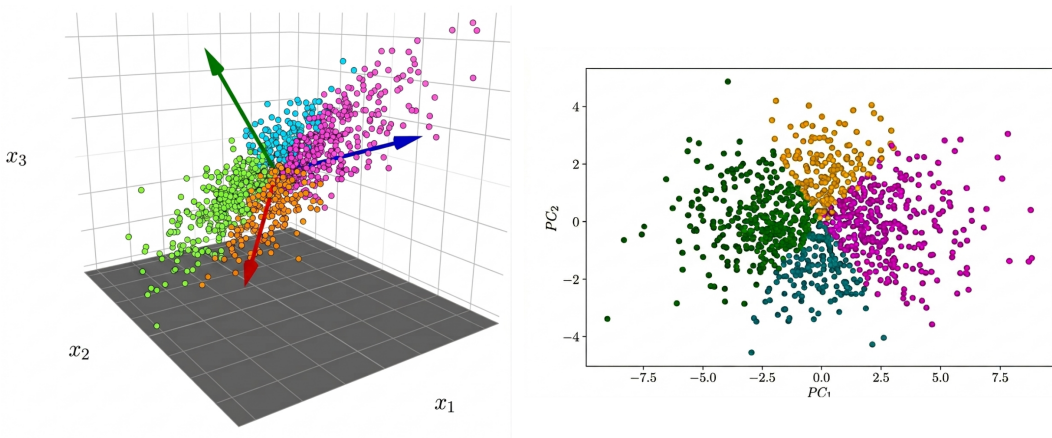


Figure 1: Dimensionality reduction visualization: Transformation from an original 3D space ( $x_1, x_2, x_3$ ) to a 2D plane defined by the principal components  $PC_1$  and  $PC_2$ .

## Principal Component Analysis - Framework

Mathematically, we consider a dataset  $\mathbf{X} \in \mathbb{R}^{m \times n}$  containing  $m$  observations and  $n$  variables (for our case,  $m = 15,000$  and  $n = 6$ ). Our goal is to transform this data into a reduced space  $\mathbb{R}^{m \times k}$  (where  $k \ll n$ ), effectively filtering out "analytical noise" to retain only the core business signal.

### STEP 1: Standardization - Scaling

To avoid variables with larger units (like USD balances) from dominating the model, we **always standardize each predictor-column**  $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$  to a mean of zero and unit variance ( $s = 1$ ):

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

**Component Breakdown:**

- $x_{ij}$  (**Raw Data**): The original value for a specific client  $i$  in variable  $j$  (e.g., the exact USD balance of client #500).
- $\bar{x}_j$  (**Variable Mean**): The average of all 15,000 observations for variable  $j$ . Subtracting this centers the data at zero.
- $s_j$  (**Standard Deviation**): The spread of variable  $j$  across the dataset. Dividing by  $s_j$  removes the original units (USD, months, etc.), creating a common scale.
- $z_{ij}$  (**Standardized Score**): The final input for PCA. A  $z = 2.0$  means the client is 2 standard deviations above the group average for that specific feature.

*Practical Note:* For manual calculations, remember that standardization simply consists of taking each data point in a column, subtracting the column's mean, and then dividing the result by the column's standard deviation.

### STEP 2: Building the Principal Components ( $Z_k$ )

We create new variables,  $\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_k$ , called **Principal Components**. Each one is a linear combination of the standardized variables ( $z_j$ ):

$$\mathbf{Z}_k = a_{k1}\mathbf{z}_1 + a_{k2}\mathbf{z}_2 + \dots + a_{kn}\mathbf{z}_n$$

**The Optimization Logic and Business Meaning:**

- **Hierarchical Capture:** These components are calculated using an optimization method where  $\mathbf{Z}_1$  captures the **maximum possible variance** (information) from the dataset. Then,  $\mathbf{Z}_2$  captures the next largest portion, and so on. This hierarchy ensures we filter the "noise" and keep only the core signal of the 15,000 clients.
- **Interpreting Magnitude (Strength):** The value of  $a_{kj}$  represents the "voice" of an original variable within the component.
  - A high absolute value (e.g.,  $|a_{kj}| > 0.7$ ) indicates that the variable is a **Dominant Driver**. For example, if  $a_{14}$  is high, then **Asset Solvency** ( $X_4$ ) is a primary pillar of the first dimension.
  - A value near zero means that variable is irrelevant for that specific behavioral profile.
- **Interpreting the Sign (Relationship):**
  - **Positive Loading (+):** The component and the variable move together. If **Liquidity** ( $X_1$ ) and **Solvency** ( $X_4$ ) both have positive loadings in  $Z_1$ , then a high  $Z_1$  score identifies a "Financially Robust" client.
  - **Negative Loading (-):** Acts as a **Contrast** or "trade-off." If **Digital Logins** ( $X_6$ ) is positive and **Credit History delays** ( $X_2$ ) is negative, the component separates customers who are digitally engaged but have no history of default.

*Strategic Insight:* The model distributes a **fixed budget of influence** among all variables within a component. This means they must "compete" for relevance: if one variable takes most of the influence, the others must take less. This competition allows the analyst to name the dimension based on the winners. If  $\mathbf{Z}_1$  is dominated by "Spending" and "Debt," it becomes the **"Credit Risk Exposure"** dimension.

### STEP 3: Eigenvalues ( $\lambda$ ) and Information Retention

Each principal component  $Z_k$  has an associated **Eigenvalue** ( $\lambda_k$ ), which represents the specific amount of variance (information) that the component has captured from the original dataset.

#### The Efficiency of Dimensional Reduction:

- **Total Variance Rule:** Because we standardized our  $n$  variables in Step 1, each variable contributes exactly 1.0 unit of variance. Therefore, the total variance in the system is always equal to the number of variables  $n$  (in our case,  $n = 6$ ).
- **The Explained Variance Ratio:** To justify how much information is retained, we use the following relationship, where the sum of all eigenvalues equals the total number of variables:

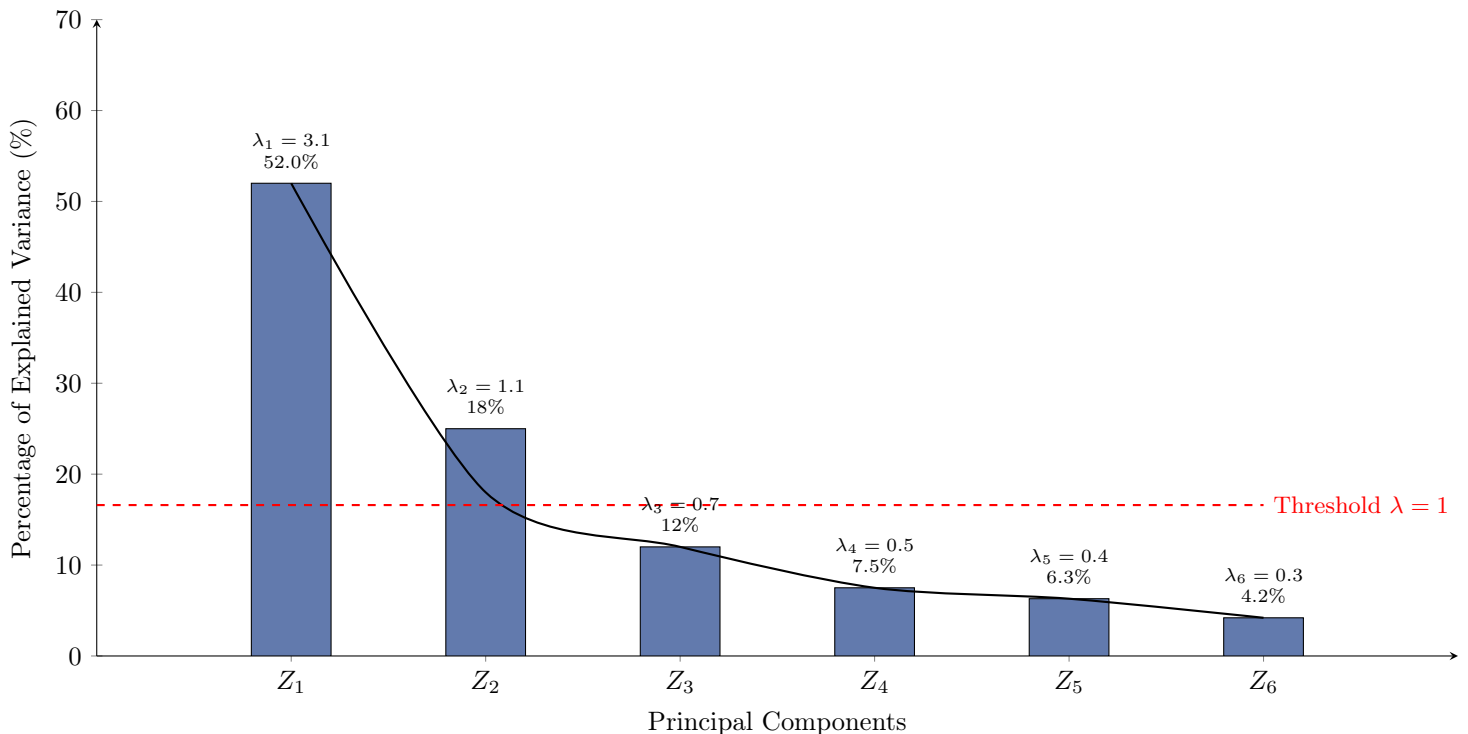
$$\text{Variance Prop.}_k = \frac{\lambda_k}{n} = \frac{\lambda_k}{\sum_{j=1}^n \lambda_j}$$

- **Cumulative Impact:** The goal is to select a number of components  $k$  such that the **Cumulative Explained Variance** is high enough (typically  $> 70\%$  or  $80\%$ ) to accurately represent the 15,000 clients without significant data loss.

*Practical Calculation for the Test:* Suppose you have  $n = 6$  variables from the Vanguard case. After running the model, you obtain  $\lambda_1 = 3.12$  and  $\lambda_2 = 1.08$ :

1.  $Z_1$  explains:  $3.12/6 = 52.0\%$  of the business signal.
2.  $Z_2$  explains:  $1.08/6 = 18.0\%$  of the business signal.
3. **Together:** These two dimensions alone capture 70% of all client behavior, allowing you to simplify the analysis significantly while losing very little insight.

*Strategic Rule of Thumb (Kaiser Criterion):* In professional practice, we typically only retain components with  $\lambda_k > 1$ . This is because a component with an eigenvalue less than 1 captures less information than a single original variable, making it inefficient for data reduction.



**Figure 2:** Scree Plot: Detailed Analysis of Eigenvalues and Explained Variance for Vanguard Capital Group.

Table 2: **Full Loading Matrix for Vanguard Capital Group**

| Variable       | $Z_1(PC_1)$ | $Z_2(PC_2)$ | $Z_3(PC_3)$ | $Z_4(PC_4)$ | $Z_5(PC_5)$ | $Z_6(PC_6)$ |
|----------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Annual Income  | 0.88        | -0.12       | 0.15        | 0.05        | -0.10       | 0.02        |
| Asset Solvency | 0.85        | -0.05       | -0.20       | 0.12        | 0.08        | -0.11       |
| Risk Score     | -0.72       | 0.45        | 0.33        | -0.08       | 0.15        | 0.04        |
| Credit Usage   | 0.15        | 0.82        | -0.11       | -0.30       | -0.05       | 0.18        |
| Debt Ratio     | -0.22       | 0.78        | 0.08        | 0.25        | -0.12       | -0.09       |
| Loyalty Index  | 0.64        | 0.31        | 0.45        | 0.10        | 0.42        | 0.35        |
| Interpretation | Wealth      | Credit      | Mix         | Resid.      | Resid.      | Resid.      |

STEP 4: Information Synthesis and Index Interpretation

- PCA as a Dimensionality Reduction Tool:** By applying PCA, we have successfully compressed the original 6-variable dataset into just 2 significant indices ( $Z_1$  and  $Z_2$ ). Together, these two components capture **70% of the total variance** ( $52\% + 18\% = 70\%$ ). This high percentage indicates that we can explain the majority of client behavior using only 1/3 of the original variables, drastically simplifying the analysis without significant information loss.
- Understanding Eigenvalues ( $\lambda$ ):**
  - Magnitude:** An eigenvalue represents the amount of variance "extracted" by a component. Since our original variables are standardized (variance = 1), a component with  $\lambda = 3.1$  is effectively doing the work of **3.1 variables**.
  - The Elbow Criterion (Scree Plot):** In the plot, we observe a sharp drop after  $Z_2$ . This "elbow" confirms that components  $Z_3$  through  $Z_6$  contribute less than a single unit of information ( $\lambda < 1$ ), qualifying them as statistical "noise".
- Core Properties of Principal Components:**
  - Orthogonality:** PCs are strictly **uncorrelated** between each other ( $r = 0$ ). This is vital for BI models because it avoids **multicollinearity**, ensuring that the Wealth Index and the Credit Index provide completely independent insights.
  - Variance Ranking:**  $Z_1$  is the "Global Dimension" (largest variance), while  $Z_2$  captures the most important remaining details.
- Advanced Correlation Analysis ( $\rho_{X_i, Z_k} = Loading_{i,k} \times \sqrt{\lambda_k}$ ):**
  - How to read the Matrix:** To find the  $Loading_{i,k}$ , locate the intersection between the **Variable** ( $X_i$ ) in the rows and the **Component** ( $Z_k$ ) in the columns of the Loading Matrix.
  - Annual Income vs.  $Z_1$ :**  $\rho = 0.88 \times \sqrt{3.1} = 0.88 \times 1.76 = \mathbf{1.55}$ . This confirms  $Z_1$  as the primary driver of financial capacity.
  - Risk Score vs.  $Z_1$ :**  $\rho = -0.72 \times \sqrt{3.1} = -0.72 \times 1.76 = \mathbf{-1.27}$ . This **inverse relationship** proves that as wealth grows, individual risk significantly drops.
  - Debt Ratio vs.  $Z_2$ :**  $\rho = 0.78 \times \sqrt{1.1} = 0.78 \times 1.05 = \mathbf{0.82}$ . This validates  $Z_2$  as a **Credit/Leverage Index**.
  - Loyalty Index vs.  $Z_1$ :**  $\rho = 0.64 \times \sqrt{3.1} = 0.64 \times 1.76 = \mathbf{1.13}$ . High wealth correlates with higher brand retention.
- Strategic Utility and Segmentation:** A BI manager can project 15,000 clients onto a 2D map ( $Z_1$  vs.  $Z_2$ ). This allows the bank to target specific quadrants:
  - Quadrant A (High  $Z_1$ , Low  $Z_2$ ):** Premium clients with high liquidity and low debt (Ideal for investment products).
  - Quadrant B (Moderate  $Z_1$ , High  $Z_2$ ):** Clients with high credit usage (Target for credit limit increases or balance transfers).

**IMPORTANT ALERT:** Students are **NOT** required to perform manual calculations to obtain loadings or eigenvalues. The evaluation focuses on your ability to extract values from the model's output, perform basic correlation arithmetic, and interpret the results to drive business strategies.

# PCA Assessment: Concept & Strategy

## 1. Algorithm Categorization

What type of algorithm is Principal Component Analysis (PCA) in the context of data science?

- A) Clustering
- B) Regression
- C) **Dimensionality Reduction**
- D) Classification

**Solution:** PCA is a fundamental algorithm used to transform a high-dimensional space into a lower-dimensional representation while preserving the maximum amount of variance.

## 2. Machine Learning Taxonomy

In terms of machine learning taxonomy, how is PCA classified?

- A) Supervised Learning
- B) **Unsupervised Learning**
- C) Reinforcement Learning
- D) Semi-supervised Learning

**Solution:** PCA is an unsupervised machine learning algorithm because it analyzes the internal structure of the data without requiring target labels or categories.

## 3. Data Standardization

Why must variables like "Annual Income" and "Credit History" be standardized before running PCA?

- A) To ensure the sample size  $m$  reaches 15,000.
- B) **To prevent variables with larger units from dominating the model variance.**
- C) To convert quantitative data into qualitative labels.
- D) To make the mean of the dataset equal to the total number of variables.

**Solution:** Standardization removes original units (USD, months, etc.) and creates a common scale ( $\mu = 0, s = 1$ ), ensuring the model treats all features equally based on information, not magnitude.

## 4. Total Variance Rule

In the Vanguard Capital Group case ( $n = 6$  variables), what is the total variance of the system after standardization?

- A) 1.0
- B) **6.0**
- C) 100%
- D) 15,000

**Solution:** Because each standardized variable contributes exactly 1.0 unit of variance, the total variance is always equal to the number of variables  $n$ .

## 5. Component Selection

According to the Kaiser Criterion, which components should be retained for business analysis?

- A) Components with eigenvalues ( $\lambda$ ) less than 1.
- B) Components that only explain demographic data.
- C) **Components with eigenvalues ( $\lambda$ ) greater than 1.**
- D) All components, to avoid losing any statistical noise.

**Solution:** A component with an eigenvalue less than 1 captures less information than a single original variable, making it inefficient for dimensionality reduction.

## 6. The Curse of Dimensionality

How does reducing dimensions help improve a machine learning model's performance?

- A) It increases the complexity of the model to capture noise.
- B) **It reduces the risk of overfitting by focusing on the core signal.**
- C) It allows the model to memorize specific training data points.
- D) It adds more variables to ensure strategic complexity.

**Solution:** Reducing dimensions acts as a regularizer, forcing the model to ignore redundant correlations and noise, which leads to better generalization on testing sets.

## 7. Interpreting Loadings

What does a high absolute loading value (e.g.,  $|a_{kj}| > 0.7$ ) indicate about a variable?

- A) The variable is irrelevant to that specific component.
- B) The variable represents a temporary market fluctuation.
- C) **The variable is a Dominant Driver within that dimension.**
- D) The variable has a mean of zero.

**Solution:** High loading values represent the "voice" or strength of an original variable within a component, identifying it as a primary pillar of that dimension.

## 8. Interpreting Contrast

If  $Z_1$  has a positive loading for "Liquidity" and a negative loading for "Risk Score," what does a high  $Z_1$  score represent?

- A) A client with high risk and low liquidity.
- B) **A financially robust client with low risk.**
- C) A client that has been excluded from the model.
- D) A client with no credit history.

**Solution:** Negative loadings act as a contrast; a high component score implies high values for positive-loading variables and low values for negative-loading variables.

## 9. Component Orthogonality

Why is it strategically useful that Principal Components are strictly uncorrelated ( $r = 0$ )?

- A) It ensures all variables have the same units.
- B) **It avoids multicollinearity and ensures independent business insights.**
- C) It makes the eigenvalues equal to each other.
- D) It prevents the creation of a Scree Plot.

**Solution:** Orthogonality ensures that dimensions like "Wealth Index" and "Credit Index" provide completely separate insights, making BI models more robust.

## 10. Strategic Utility

In the Vanguard case, if  $Z_1$  and  $Z_2$  capture 70% of the variance, what is the primary benefit for the BI analyst?

- A) The analyst must now find 4 more variables to reach 100%.
- B) **The analyst can simplify the analysis to 2 dimensions without significant data loss.**
- C) The analyst can ignore repayment behavior entirely.
- D) The analyst should stop using visualization tools.

**Solution:** This reduction captures the majority of client behavior using only 1/3 of the original variables, facilitating data storytelling and executive decision-making.